

Exam Date & Time: 06-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. EXAMINATIONS - JANUARY 2024
SUBJECT: ECE 3151-ANALOG AND DIGITAL COMMUNICATION
ANALOG AND DIGITAL COMMUNICATION [ECE 3151]

Marks: 50

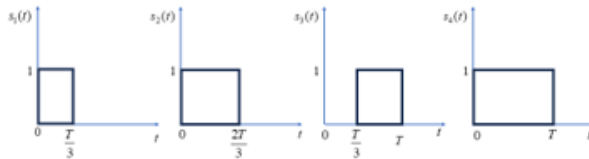
Duration: 180 mins.

A

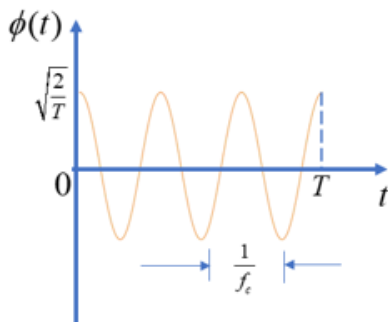
Answer all the questions.

Missing data may be suitably assumed.

- 1A) Explain the generation method of amplitude modulated (AM) signal using the switching modulator with relevant expression, neat circuit diagram, and waveforms. Also, explain the envelope detection method with its circuit diagram and waveform representation. (5)
- 1B) An angle modulated signal is defined by $s(t) = 30 \cos[2\pi \times 10^6 t + 0.5 \sin 2000\pi t]$ V.
Find
 i) Power in the modulated signal (3)
 ii) Frequency deviation Δf
 iii) Approximate transmission bandwidth
- 1C) Obtain Fourier transform of signal $e^{-at} u(t)$. (2)
- 2A) Consider the signal $s_1(t)$, $s_2(t)$, $s_3(t)$ and $s_4(t)$ shown in the Figure. Use the Gram-Schmidt orthogonalization procedure to find a set of orthonormal basis functions. (5)



- 2B) Consider a radio frequency (RF) pulse of duration T seconds and unit energy as given by the following figure. (3)



- i) Find the impulse response of the matched filter to the given signal.
 ii) Find the filter output and plot its curve.

- 2C) (2)

Assume that three signals $s_1(t)$, $s_2(t)$ and $s_3(t)$ can be represented using two orthonormal basis functions $\phi_1(t)$ and $\phi_2(t)$. The coordinates of these signals are $s_1 = (3, 3)$, $s_2 = (-2, -2)$ and $s_3 = (-3, -3)$. Draw the constellation diagram and express the three signals $s_1(t)$, $s_2(t)$ and $s_3(t)$ as a linear combination of the given basis functions.

- 3A) Consider the binary data sequence **1 1 1 1 1 0 1** which is applied to the input of a duo binary system.
 i) Determine the output of the duo-binary coder and the corresponding receiver output, without a precoder.
 ii) Determine the output of the duo-binary coder and the corresponding receiver output, with a precoder. (5)

Assume reference bit to be 1.

- 3B) With a block diagram and relevant mathematical expressions, explain the working of adaptive delta modulation scheme. (3)

- 3C) For the given sequence "**10011011**", plot the resulting waveform, assuming that following PAM format is used. (2)
 i) NRZ unipolar
 ii) NRZ bipolar

- 4A) Starting from fundamentals, with the help neat block diagram of receiver for baseband transmission of binary encoded PCM wave, derive the expression for average probability of error for binary amplitude shift keying modulation (PCM ON-OFF signalling) scheme. (5)

- 4B) Consider transmission of a dibit **00** using QPSK modulation, where the conditional probability of joint event is $x_1 < 0$ and $x_2 < 0$. Determine average probability of symbol error with all intermediate steps. Assume AWGN channel. (3)

- 4C) The binary data sequence **1 0 1 1 1 1 0 1** is applied to the input of a DPSK transmitter. Determine the differentially encoded sequence and transmitted phase. **Assume initial reference bit as 1.** (2)

- 5A) Construct minimum variance Huffman code for the given source probabilities $p_k = [0.05, 0.15, 0.2, 0.38, 0.22]$. Also determine the average code length of this Huffman code. (5)

- 5B) Determine the entropy, efficiency and redundancy of the Huffman code as given in Q5A. (3)

- 5C) Determine the minimum variance of this Huffman code as given in Q5A. (2)

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